

Artifact: Certified Geometric Robustness - Super-DEEPG

Super-DEEPG verifies the robustness of neural networks against geometric perturbations. For instance, if a safety requirement specifies that a neural network must be robust to rotation angles between -30° and 30° , Super-DEEPG can verify the percentage of images in a dataset that satisfy this property.

1 Requirements and setup

1.1 Installation

The artifact requires **Python ≥ 3.11** . To set up the environment and install the necessary dependencies, please execute the following commands in your terminal:

```
1 unzip SuperDEEPG_artifact.zip
2 cd SuperDEEPG_artifact/
3 python3 -m venv .venv
4 source .venv/bin/activate
5 git clone https://github.com/Verified-Intelligence/auto_LiRPA
6 cd auto_LiRPA
7 pip install .
8 cd ..
9 pip install -r requirements.txt
```

Datasets/Models: MNIST and CIFAR10 are handled automatically. TinyImageNet and specific model weights must be placed in `experiment/dataset/` and `experiment/model/` as specified in the README.

Hardware: The tool is optimized for GPU parallelization to handle pixel-wise and sub-interval computations.

1.2 Sanity check (Quick start)

To validate the setup, run:

```
1 python main_batch.py --config_path experiment/specification/example.yaml
2 python analyze.py experiment/results/example_results.csv
```

Expected output: The analysis should report **98% certified images** for MNIST.

2 Expected reproducibility outcomes

2.1 Metrics

As described in the paper, we evaluate the performance using two primary metrics:

- **Certified accuracy (%)**: The percentage of test samples guaranteed to be correctly classified under the specified geometric transformation range.
- **Certification time**: The total duration required to certify each image (seconds per image).

3 Mapping paper outcomes to artifact

The mapping between the paper’s Table results and the artifact is established via specific execution scripts located in the `scripts/` directory. Each script runs the verification process and automatically displays the certified accuracy and certification time. To reproduce any result, follow these steps:

- Choose the script corresponding to the desired result in Table 1.
- Run the script from the root of the artifact:

```
bash scripts/TableX-Dataset-Transfo-IS.bash
```

Table	Dataset	Transformation	IS	Reproduction Script
Table 2	MNIST	Rotation [-30°, 30°]	1.0	scripts/Table2-MNIST-R30-1.0.bash
			6.0	scripts/Table2-MNIST-R30-6.0a.bash scripts/Table2-MNIST-R30-6.0b.bash
		Scaling [-2, 2]	0.04	scripts/Table2-MNIST-Sc2-0.04.bash
		Shearing [-2, 2]	0.04	scripts/Table2-MNIST-Sh2-0.04.bash
		Translation [-1, 1]	0.2	scripts/Table2-MNIST-T1-0.2.bash
	CIFAR10	Rotation [-10°, 10°]	1.0	scripts/Table2-CIFAR10-R10-1.0a.bash scripts/Table2-CIFAR10-R10-1.0b.bash
			1.0	scripts/Table2-CIFAR10-Sc1-1.0.bash
		Scaling [-1, 1]	1.0	scripts/Table2-CIFAR10-Sh2-0.02a.bash scripts/Table2-CIFAR10-Sh2-0.02b.bash
		Shearing [-2, 2]	0.02	
Table 3	MNIST	Rotation [-30°, 30°]	2.0	scripts/Table3-MNIST-R30-2.0.bash
	CIFAR10	Rotation [-10°, 10°]	0.01	scripts/Table3-CIFAR10-R10-0.01.bash
		Shearing [-0.02, 0.02]	0.001	scripts/Table3-TinyImgNet-Sh0.02-0.001.bash
	TinyImgNet	Scaling [0.98, 1.02]	0.001	scripts/Table3-TinyImgNet-Sc0.02-0.001.bash
		Rotation [-5°, 5°]	0.08	scripts/Table3-TinyImgNet-R5-0.08.bash
Table 6	TinyImgNet	Rotation [-5°, 5°]	0.04	scripts/Table6-TinyImgNet-R5-0.04.bash
		Scaling [0.98, 1.02]	0.0002	scripts/Table6-TinyImgNet-Sc0.02-0.0002.bash
		Shearing [-0.02, 0.02]	0.0003	scripts/Table6-TinyImgNet-Sh0.02-0.0003.bash

Table 1: This table links the experimental results to their respective bash scripts in `scripts/`. Each script handles both the verification and the analysis of results.

4 Additional information

The README file provides requirements and setup instructions. It outlines the codebase structure and defines the role of each directory. Additionally, the documentation includes an explanation of all YAML hyperparameters. Users can adapt the tool to verify their own neural networks. To do so:

1. Place the new model weights (`.pt` or `.onnx`) in `experiment/model/`.
2. Create a new YAML configuration in `experiment/specification/` specifying the transformation range (e.g., `["rotate", -10, 10, 1]`).
3. Execute `main_batch.py` with the new configuration path.